

**MASTER OF COMPUTER
APPLICATIONS/BACHELOR OF
COMPUTER APPLICATIONS
(REVISED) (MCA/BCA)**

Term-End Examination

June, 2023

**MCS-011 : PROBLEM SOLVING AND
PROGRAMMING**

Time : 3 Hours

Maximum Marks : 100

Weightage : 75%

***Note : Question No. 1 is compulsory. Answer any
three questions from the rest.***

1. (a) Write an algorithm to calculate simple interest. Also draw flow chart for this algorithm (**Hint** : $SI = (PTR)/100$). 10
- (b) Write a C program to create a structure to store name, rollnumber, address and course (BCA, MCA) of ten students. Use array of structure to display details of the students.

10

- (c) Write a C program which read an array of ten integer values and search a given value in the array. If that value exist in the array display its square otherwise display. "The value is missing." 10
- (d) Write a program in 'C' using printers to find the length of a given string. 10
2. (a) Write a C program to find the sum of the following series upto 20 terms : 6
- 1 + 4 + 7 + 10 + 13
- (b) Briefly explain *two* categories of constants in C. 4
- (c) Write a C program for multiplication of two 4×4 matrices. 10
3. (a) Explain the difference between `a++` and `++a`. Also find the value that would be printed by the following code : 6
- ```
int a = 4;
int b = 2;
int c = 0;

a = c++;
a = a + b++;
printf("a = %d", a);
```

- (b) Explain the use of conditional operator with the help of an example. 4
- (c) What is for loop ? How is it different from do ..... while loop ? Explain with the help of an example. Also explain use of “break statement”. 10
4. (a) Write a ‘C’ program, using structures, to calculate the gross salary and net salary if Basic, D. A., T. A. and deduction are given as input. 10
- (b) Write the syntax and explain the use of the following functions in C :  $2 \times 4 = 8$
- (i) return
  - (ii) malloc
  - (iii) isupper
  - (iv) fclose
- (c) Define a macro to find the cube of a given number. 2
5. (a) Write C program which take a string as input and print it in reverse order, without using any in-built function. 5

- (b) Write a C program to check whether a file whose name is given exist or not in the given list of file names. 5
- (c) Explain the following with the help of a suitable example for each :  $4 \times 2\frac{1}{2} = 10$
- (i) File access modes
  - (ii) Array of pointers
  - (iii) Automatic variable
  - (iv) Size of operator